

MAT11-4

Discrete Mathematics

Proof, discrete structures, and a bridge to stochastic reasoning

Learn to prove what is true, disprove what is false, and reason — case by case — about the discrete world.

Albert School — 22 hours

Preface

In **Foundations** (MAT11-1) you learned the vocabulary of mathematics: how to read a quantified statement, how to negate an implication, how to use set notation, and how to manipulate functions, sequences and sums. You learned, in effect, to **speak** mathematics. This course teaches you to **argue** in it.

The difference is real. Knowing that “ $P \Rightarrow Q$ ” is logically equivalent to its contrapositive “ $\neg Q \Rightarrow \neg P$ ” is a fact about language; turning that fact into a complete, convincing proof of a theorem is a craft. The first third of this book is devoted to that craft — direct proof, contrapositive, contradiction, and the two faces of induction — together with its indispensable companion, the art of **disproof** by counterexample. A statement is not a theorem until you can prove it, and not worth proving until you have checked it is not false.

With those tools in hand we turn them on the discrete structures that the rest of the AD track depends on. **Equivalence relations** and **partitions** are the same idea seen from two sides: a way of splitting a set into clean, non-overlapping pieces. **Recurrences** define sequences step by step, and induction is exactly the instrument for proving the closed forms they hide. **Divisibility** and **modular arithmetic** give us a finite, cyclic number system where the Euclidean algorithm and Bézout’s identity decide what can and cannot be solved.

Finally — and this is the quiet agenda of the whole course — we build a **bridge to probability**. Long before you meet a probability axiom in **Stochastic Reasoning** (MAT21-4), you will learn to decompose a computation over a partition (“split into cases, weight each case, recombine”), to read a tree diagram of sequential conditions, and to model a finite system that evolves in time as a matrix you can raise to a power. These are the law of total expectation and the Markov chain, met here as concrete counting and linear algebra, stripped of their probabilistic clothing. When the axioms arrive next semester, the reasoning will already feel like home.

One theme recurs throughout: **partition-and-condition**. Break a complicated object into disjoint pieces, understand each piece, recombine. It is how equivalence classes are built, how strong induction splits a problem, how the Euclidean algorithm shrinks a pair of integers, and how a probability is computed by conditioning. Keep it in view and the course will feel like one idea told many ways.

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1 The anatomy of a mathematical statement

Before you can prove a theorem you must know **exactly** what it claims. Most failed proofs are not failures of cleverness but of reading: the student set out to prove something subtly different from what was written. This chapter is about reading a statement so precisely that the shape of its proof becomes almost forced.

1.1 A statement is a promise about every case

Consider the claim

“For every integer n , if n is odd then n^2 is odd.”

Read it slowly. It is not a statement about one number; it is a **promise** that covers $n = 1, 3, 5, \dots$ and $n = -7$ and every other odd integer at once. To **prove** it you must argue for an arbitrary odd n without ever choosing a specific value. To **disprove** it — were it false — you would need only a single n that breaks it. This asymmetry between proof and disproof is the engine of the whole chapter, and it comes straight from the quantifiers.

1.2 Quantifiers and their order

You met $\forall$ (“for all”) and $\exists$ (“there exists”) in MAT11-1. Here we take their **interaction** seriously, because reversing two quantifiers usually changes the meaning entirely.

Definition 1 (Quantifier order) In $\forall x, \exists y, P(x, y)$ the choice of y is allowed to **depend on x** : for each x we may pick a different y . In $\exists y, \forall x, P(x, y)$ a **single y** must work for **every x** simultaneously. The second is strictly stronger.

Worked example — Why order is not a formality. Let $P(x, y)$ be “ $y > x$ ” over the integers.

- $\forall x, \exists y, (y > x)$ is **true**: given any x , take $y = x + 1$.
- $\exists y, \forall x, (y > x)$ is **false**: no single integer exceeds every integer.

Same predicate, opposite truth values — the only thing that changed was the order of the quantifiers.

Common pitfall The most common reading error is to treat “ $\forall x, \exists y$ ” as if one y were promised in advance. The convergence definition you saw in MAT11-1 — $\forall \varepsilon > 0, \exists N, \forall n \geq N, |u_n - \ell| < \varepsilon$ — is exactly this trap: the threshold N is allowed to depend on ε , and almost always must. Reading it as “one N for all ε ” describes a different (and false) statement.

1.3 Negation: turning a claim inside out

To disprove a statement you negate it, and the negation tells you precisely what a counterexample must look like. The rules push the negation **inward**, flipping each quantifier and each connective as it passes.

Definition 2 (Negation rules)

$$\begin{aligned}\neg(\forall x, P(x)) &\Leftrightarrow \exists x, \neg P(x), & \neg(\exists x, P(x)) &\Leftrightarrow \forall x, \neg P(x), \\ \neg(P \wedge Q) &\Leftrightarrow (\neg P) \vee (\neg Q), & \neg(P \vee Q) &\Leftrightarrow (\neg P) \wedge (\neg Q), \\ \neg(P \Rightarrow Q) &\Leftrightarrow P \wedge (\neg Q).\end{aligned}$$

The last rule deserves emphasis: the negation of “ $P \Rightarrow Q$ ” is **not** another implication. It is the concrete assertion “ P holds and yet Q fails” — which is exactly a counterexample.

Worked example — Negating a convergence statement. Negate $\forall \varepsilon > 0, \exists N, \forall n \geq N, |u_n - \ell| < \varepsilon$. Flip each quantifier in turn and negate the inner inequality:

$$\exists \varepsilon > 0, \forall N, \exists n \geq N, |u_n - \ell| \geq \varepsilon.$$

In words: there is a fixed tolerance ε that the sequence keeps violating — no matter how far out you look ($\forall N$), some later index $n \geq N$ still lands at least ε away. That is precisely what “ u_n does **not** converge to ℓ ” means.

Exercises

1. Write the contrapositive and the negation of: “if n^2 is even then n is even.” Which of the three statements (original, contrapositive, negation) are true?
2. State the difference in meaning between $\forall x, \exists y, (xy = 1)$ and $\exists y, \forall x, (xy = 1)$ over the nonzero reals, and give the truth value of each.
3. Negate: “every continuous function on $[0, 1]$ attains a maximum.” What would a counterexample have to be?

2 Methods of proof

We now build the four standard proof techniques. For each, the strategy is dictated by the **logical shape** of the goal you read off in Chapter 1.

2.1 Direct proof

To prove $P \Rightarrow Q$ directly: assume P , and reason forward to Q . This is the default; reach for the others only when the direct route is awkward.

Worked example — Odd squared is odd, directly. Claim. If n is odd then n^2 is odd.
Proof. Assume n is odd, so $n = 2k + 1$ for some integer k . Then

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

which is $2m + 1$ with $m = 2k^2 + 2k \in \mathbb{Z}$. Hence n^2 is odd.

2.2 Proof by contrapositive

Because $P \Rightarrow Q$ and $\neg Q \Rightarrow \neg P$ are logically equivalent, you may prove either. Choose the contrapositive when the negated conclusion gives you something more concrete to work with than the original hypothesis.

Worked example — When the contrapositive is easier. Claim. If n^2 is even then n is even. A direct proof would start from “ n^2 is even” — awkward, since it tells us little about n itself. The contrapositive is “if n is odd then n^2 is odd,” which we just proved directly. Done.

Common pitfall The contrapositive of $P \Rightarrow Q$ is $\neg Q \Rightarrow \neg P$, **not** the **converse** $Q \Rightarrow P$ nor the **inverse** $\neg P \Rightarrow \neg Q$. The converse and inverse are generally **not** equivalent to the original. “If it rains, the ground is wet” does not give “if the ground is wet, it rained.”

2.3 Proof by contradiction

To prove a statement S by contradiction, assume $\neg S$ and derive an impossibility; since $\neg S$ leads to nonsense, S must hold. For an implication $P \Rightarrow Q$ this means assuming $P \wedge \neg Q$ (the negation we computed above) and reaching a contradiction.

Worked example — Irrationality of $\sqrt{2}$. **Claim.** $\sqrt{2}$ is irrational. **Proof.** Suppose not: $\sqrt{2} = p/q$ with p, q integers sharing no common factor (lowest terms). Squaring, $p^2 = 2q^2$, so p^2 is even, hence p is even (Chapter 2's lemma), say $p = 2k$. Then $4k^2 = 2q^2$, i.e. $q^2 = 2k^2$, so q is even too. But then p and q share the factor 2, contradicting “lowest terms.” Therefore no such fraction exists and $\sqrt{2}$ is irrational.

Common pitfall Contradiction is powerful but easy to abuse. If your “contradiction” is really just a **direct** derivation of Q that never used the assumption $\neg Q$, you have written a direct proof wearing a disguise — drop the disguise. Genuine contradiction proofs use the extra hypothesis $\neg Q$ essentially.

2.4 Mathematical induction

Induction proves a statement $P(n)$ for **all** integers n from some starting point. It is the engine for everything recursive.

2.4.1 A concrete tower first

Suppose you want $1 + 2 + \dots + n = n(n+1)/2$. You could verify $n = 1, 2, 3, \dots$, but no finite amount of checking covers **all** n . Induction replaces infinite checking with two finite tasks: confirm the first case, and show that **each** case forces the next. Like dominoes, if the first falls and each knocks over its successor, all fall.

Theorem 3 (Weak (ordinary) induction) Let $P(n)$ be a statement defined for integers $n \geq n_0$. If

1. (**base case**) $P(n_0)$ is true, and
2. (**inductive step**) $P(n) \Rightarrow P(n+1)$ for every $n \geq n_0$,

then $P(n)$ is true for all $n \geq n_0$.

Worked example — The Gauss sum. **Claim.** $\sum_{k=1}^n k = n(n+1)/2$ for all $n \geq 1$. **Base.** $n = 1$: left side 1, right side $1 \cdot 2/2 = 1$. ✓ **Step.** Assume $\sum_{k=1}^n k = n(n+1)/2$ (the **induction hypothesis**). Then

$$\sum_{k=1}^{n+1} k = \underbrace{\sum_{k=1}^n k}_{= \frac{n(n+1)}{2}} + (n+1) = (n+1) \left(\frac{n}{2} + 1 \right) = (n+1) \frac{n+2}{2},$$

which is the formula with $n+1$ in place of n . By induction the formula holds for all $n \geq 1$.

Common pitfall Two failures recur. (1) **Forgetting the base case.** Without it the dominoes may all be primed yet none ever falls; one can “prove” the false statement $n = n+1$ if only the step is checked. (2) **Assuming what you want to prove.** The induction

hypothesis is $P(n)$, the single previous case — not $P(n+1)$, the thing you are trying to establish. Using $P(n+1)$ inside the step is circular.

2.4.2 When one previous case is not enough

Sometimes the step needs **several** earlier cases, or **all** of them. Strong induction grants exactly that.

Theorem 4 (Strong induction) Let $P(n)$ be defined for $n \geq n_0$. If $P(n_0)$ holds and, for every $n \geq n_0$,

$$(P(n_0) \wedge P(n_0 + 1) \wedge \dots \wedge P(n)) \Rightarrow P(n + 1),$$

then $P(n)$ holds for all $n \geq n_0$. (One may need to verify a few base cases at the start, enough to feed the first genuine step.)

Worked example — Every integer $n \geq 2$ has a prime factor. Claim. Every integer $n \geq 2$ is a product of primes. **Proof (strong induction).** Base: 2 is prime. Step: take $n \geq 2$ and assume the claim for all $2 \leq m \leq n$. If $n+1$ is prime we are done. Otherwise $n+1 = ab$ with $2 \leq a, b \leq n$; by the **strong** hypothesis both a and b factor into primes, and concatenating those factorizations gives one for $n+1$. Weak induction would not suffice here: a and b are generally far below n , not the immediately preceding case.

Exercises

4. Prove by induction that $\sum_{k=1}^n k^2 = n(n+1)(2n+1)/6$.
5. Prove $2^n > n^2$ for all $n \geq 5$. Identify where the base case must start and why $n = 5$, not $n = 1$.
6. Prove that every amount of postage of 8 cents or more can be made from 3-cent and 5-cent stamps. (Strong induction: assume the result down to $n - 3$.)
7. Find the flaw in this “proof” that all horses are the same colour: “any one horse is the same colour as itself; assume any n horses are the same colour; given $n + 1$ horses, the first n share a colour and the last n share a colour, so all $n + 1$ do.” Where exactly does the step break?

3 Counterexamples and failure modes

Proof’s twin is disproof. A **universal** claim $\forall x, P(x)$ is destroyed by a single x with $\neg P(x)$ — that is what the negation rule tells us. Learning to hunt counterexamples is a skill in its own right, and it sharpens your sense of **which hypotheses a theorem actually needs**.

3.1 One witness is enough

Worked example — Disproving a universal. Claim. “Every odd number is prime.” **Disproof.** 9 is odd but $9 = 3 \times 3$ is not prime. A single witness finishes it; we need not say anything about other numbers.

3.2 Finding the missing hypothesis

Often a plausible claim is **almost** true — it fails only on a boundary or under a missing assumption. The instructive move is to locate the exact hypothesis that repairs it.

Worked example — Repairing a false claim. Claim. “ $x^2 > x$ for all real x .” **Disproof.** It fails on $[0, 1]$: at $x = 1/2$, $x^2 = 1/4 < 1/2$. Worse, $x^2 = x$ at $x = 0$ and $x = 1$. The claim is true exactly when $x < 0$ or $x > 1$; the missing hypothesis is “ $x \notin [0, 1]$.” This is **boundary-condition reasoning**: the interesting behaviour lives at the edges 0 and 1 where the two sides cross.

Common pitfall A counterexample must satisfy **all** the stated hypotheses and **violate the conclusion**. A “counterexample” that quietly drops a hypothesis disproves nothing — it merely shows the hypothesis was needed. Always check your witness against every clause of the claim.

Exercises

8. Disprove “if $a \mid bc$ then $a \mid b$ or $a \mid c$ ” with an explicit counterexample, then state the extra hypothesis on a that makes it true.
9. The claim “a sum of two irrational numbers is irrational” is false. Give a counterexample and identify the structural reason it fails.
10. Find a function that is continuous on the open interval $(0, 1)$ but attains no maximum there, and say which hypothesis of the extreme value theorem is missing.

4 Equivalence relations and partitions

We now meet the first discrete structure, and the most important one for what follows: a clean way to **split a set into pieces**. The same idea wears two faces — a **relation** that groups elements, and a **partition** that lists the groups — and the central theorem is that the two faces are the same.

4.1 From “sameness” to a relation

When we say two integers are “the same modulo 3,” or two students are “in the same class,” we are declaring some elements **equivalent** without declaring them **equal**. The three properties below are exactly what any reasonable notion of sameness must satisfy.

Definition 5 (Equivalence relation) A relation $\sim$ on a set S is an **equivalence relation** if it is

- **reflexive**: $a \sim a$ for every $a \in S$;
- **symmetric**: $a \sim b$ implies $b \sim a$;
- **transitive**: $a \sim b$ and $b \sim c$ imply $a \sim c$.

Worked example — Three relations, only some equivalences.

- “Has the same remainder mod 3” on $\mathbb{Z}$: reflexive, symmetric, transitive — an equivalence relation.
- “ $\leq$ ” on $\mathbb{R}$: reflexive and transitive but **not** symmetric ($2 \leq 3$ yet $3 \not\leq 2$). Not an equivalence relation.
- “Differs by exactly 1” on $\mathbb{Z}$: symmetric but neither reflexive (a does not differ from itself) nor transitive. Not an equivalence relation.

4.2 Equivalence classes and the partition they form

Definition 6 (Equivalence class) The **equivalence class** of $a \in S$ is

$$[a] = \{x \in S : x \sim a\},$$

the set of all elements equivalent to a . Any element of a class is a **representative** of it.

Definition 7 (Partition) A **partition** of S is a collection of non-empty subsets (the **blocks**) that are pairwise disjoint and whose union is all of S : every element of S lies in exactly one block.

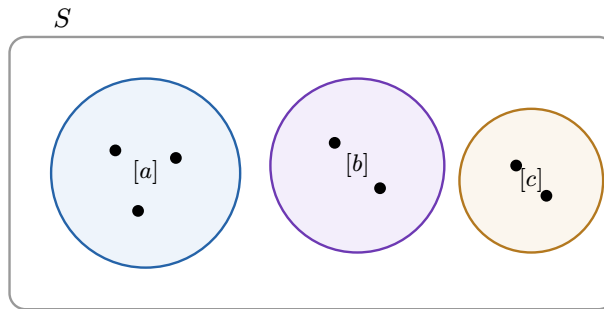


Figure 1: The blocks of a partition tile the set S with no overlaps and no gaps. The classes $[a], [b], [c]$ of an equivalence relation are exactly such blocks — this is the content of the theorem below.

Theorem 8 (Equivalence relations $\Leftrightarrow$ partitions) Let $\sim$ be an equivalence relation on S . Then its equivalence classes form a partition of S : each a lies in $[a]$ (reflexivity), and two classes are either identical or disjoint. Conversely, every partition of S arises from a unique equivalence relation — declare $a \sim b$ iff a and b lie in the same block. The set of classes is the **quotient set** $S/\sim$.

Proof. Reflexivity gives $a \in [a]$, so the classes cover S . Suppose $[a]$ and $[b]$ share an element x : then $x \sim a$ and $x \sim b$, so by symmetry and transitivity $a \sim b$, whence any $y \sim a$ also satisfies $y \sim b$ and vice versa — that is, $[a] = [b]$. So distinct classes are disjoint. The converse is a direct check that “same block” is reflexive, symmetric and transitive. $\square$

Remark This theorem is why partitions matter for the rest of the course. Whenever you decompose a set “by cases” — residues mod n , outcomes of a first coin flip, states of a system — you are choosing a partition, and the partition-and-condition reasoning of Chapter 9 runs on exactly this structure.

Worked example — Congruence mod n . On $\mathbb{Z}$, “ $a \sim b$ iff $n \mid (a - b)$ ” is an equivalence relation whose classes are the n **residue classes** $[0], [1], \dots, [n - 1]$. They partition $\mathbb{Z}$, and the quotient set is written $\mathbb{Z}/n\mathbb{Z}$ — the object Chapter 8 turns into an arithmetic.

Exercises

11. Verify that “ $a \sim b$ iff $a - b \in \mathbb{Z}$ ” is an equivalence relation on $\mathbb{R}$, and describe a natural representative for each class in $[0, 1)$.
12. On the plane minus the origin, declare two points equivalent if they lie on the same line through the origin. Show this is an equivalence relation and describe the classes geometrically.
13. List all partitions of the set $\{1, 2, 3\}$ and write the equivalence relation corresponding to each.

5 Counting, recurrences, and the binomial theorem

Counting is where discrete reasoning meets the recursive thinking induction prepared us for. We assemble the basic counting tools, then see how the most useful combinatorial quantities satisfy **recurrences** — sequences defined step by step — and how induction proves their closed forms.

5.1 Permutations and combinations

Definition 9 (Permutations and combinations) From n distinct objects, the number of **ordered** selections of k of them is

$$P(n, k) = \frac{n!}{(n - k)!} = n(n - 1) \dots (n - k + 1),$$

and the number of **unordered** selections (subsets of size k) is the **binomial coefficient**

$$\binom{n}{k} = \frac{P(n, k)}{k!} = \frac{n!}{k!(n - k)!}.$$

Worked example — Order matters, or it does not. Choosing a president and a vice-president from 5 people: order matters, $P(5, 2) = 5 \cdot 4 = 20$. Choosing a 2-person committee from the same 5: order is irrelevant, $\binom{5}{2} = 20/2! = 10$. The factor $k! = 2$ is exactly the number of ways to order each committee.

5.2 A recurrence for the binomial coefficients

Definition 10 (Recurrence relation) A sequence is defined **recursively** by giving its first term(s) and a rule that produces each later term from earlier ones — for instance $u_0 = 1$, $u_{n+1} = 2u_n$.

The binomial coefficients themselves obey a beautiful recurrence, the arithmetic behind Pascal’s triangle.

Theorem 11 (Pascal’s rule) For $1 \leq k \leq n$,

$$\binom{n}{k} = \binom{n - 1}{k - 1} + \binom{n - 1}{k}.$$

Proof. Count the size- k subsets of $\{1, \dots, n\}$ by whether they contain the element n . Those that **do** are a size- $(k - 1)$ subset of the remaining $n - 1$ elements, counted by $\binom{n - 1}{k - 1}$; those that **do not** are a size- k subset of those $n - 1$, counted by $\binom{n - 1}{k}$. The two cases partition all such subsets, so the counts add. $\square$

Remark Notice the proof is itself a **partition-and-condition** argument: split the objects by a yes/no condition (“does it contain n ?”), count each block, recombine. The same move will compute probabilities in Chapter 9.

5.3 The binomial theorem

Theorem 12 (Binomial theorem) For any $n \in \mathbb{N}$,

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}.$$

Proof. By induction on n , using Pascal’s rule. Base $n = 0$: both sides are 1. Step: assuming the formula for n , multiply by $(x + y)$ and collect the coefficient of $x^k y^{n+1-k}$; it is $\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}$ by Pascal’s rule, which is exactly the claim for $n + 1$. $\square$

Worked example — A quick expansion. $(x + y)^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$, the coefficients 1, 4, 6, 4, 1 being row 4 of Pascal’s triangle.

Exercises

14. In how many ways can 7 books be arranged on a shelf? How many ways to choose 3 of them to take on a trip? State which is a permutation and which a combination.
15. Verify Pascal’s rule numerically for $\binom{6}{2}$, and use the binomial theorem to expand $(2a - b)^3$.
16. Prove $\sum_{k=0}^n \binom{n}{k} = 2^n$ in two ways: by setting $x = y = 1$ in the binomial theorem, and by counting all subsets of an n -element set.

6 Recurrences and stability

Having met recurrences as definitions, we study them as **processes**: given the rule, what closed form does the sequence hide, and where does it settle in the long run? The closed form is found by guessing-and-proving (induction); the long-run behaviour is read off a single parameter.

6.1 From recurrence to closed form

Worked example — Doubling. Let $u_0 = 1$ and $u_{n+1} = 2u_n$. Computing 1, 2, 4, 8, ... suggests $u_n = 2^n$. **Proof by induction.** Base $u_0 = 2^0 = 1$. Step: $u_{n+1} = 2u_n = 2 \cdot 2^n = 2^{n+1}$. The closed form is justified, not merely guessed.

6.2 Affine recurrences and the fixed point

The workhorse of the course is the **affine** (linear, first-order) recurrence $u_{n+1} = au_n + b$. Its behaviour is governed by its **fixed point** — the value that maps to itself.

Theorem 13 (Closed form of an affine recurrence) Suppose $u_{n+1} = au_n + b$ with $a \neq 1$. The **fixed point** ℓ solving $\ell = a\ell + b$ is $\ell = b/(1 - a)$, and

$$u_n - \ell = a^n(u_0 - \ell), \quad \text{so} \quad u_n = \ell + a^n(u_0 - \ell).$$

Proof. Subtract $\ell = a\ell + b$ from $u_{n+1} = au_n + b$ to get $u_{n+1} - \ell = a(u_n - \ell)$. Thus the shifted sequence $v_n = u_n - \ell$ is geometric with ratio a , so $v_n = a^n v_0$ — which is the claim. (A clean induction proof in disguise.) $\square$

6.3 Stability: where the sequence goes

Proposition 14 (Stability of an affine recurrence) With $a \neq 1$ and fixed point ℓ :

- if $|a| < 1$, then $u_n \rightarrow \ell$ — the fixed point is **attracting** and the recurrence is **stable**;
- if $|a| > 1$, then $|u_n - \ell| \rightarrow \infty$ (unless $u_0 = \ell$) — the fixed point is **repelling** and the recurrence is **unstable**;
- if $a = -1$, the sequence oscillates between two values without converging.

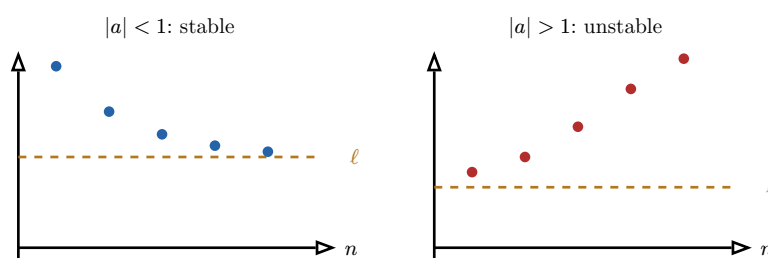


Figure 2: The single number $|a|$ decides everything. When $|a| < 1$ the iterates are pulled toward the fixed point ℓ ; when $|a| > 1$ they are pushed away. This is the discrete cousin of the stability analysis you will meet for dynamical systems and Markov chains.

Worked example — A savings account with withdrawals. Each year a balance grows by 5% then €1000 is withdrawn: $u_{n+1} = 1.05u_n - 1000$. Here $a = 1.05 > 1$, fixed point $\ell = -1000/(1 - 1.05) = 20000$. Since $|a| > 1$, balances above €20{,}000 grow without bound and balances below it are driven to $-\infty$: the threshold €20{,}000 is the repelling fixed point.

Exercises

- For $u_0 = 3$, $u_{n+1} = (1/2)u_n + 2$, find the fixed point, the closed form, and the limit. Prove the closed form by induction.
- A population satisfies $p_{n+1} = 1.1p_n - 50$. Find the fixed point and classify it as attracting or repelling. For which p_0 does the population survive?
- Show that if $a = 1$ the affine recurrence $u_{n+1} = u_n + b$ has closed form $u_n = u_0 + nb$ (an arithmetic sequence), so the fixed-point method does not apply.

7 Divisibility and the Euclidean algorithm

We now develop the arithmetic of the integers that modular arithmetic and the counting of solvable equations rest on. Everything flows from one fact you have known since childhood — division with remainder — made precise.

7.1 Euclidean division

Theorem 15 (Euclidean division) For every $a \in \mathbb{Z}$ and $b \in \mathbb{N}^*$ there exist **unique** integers q (the quotient) and r (the remainder) with

$$a = bq + r, \quad 0 \leq r < b.$$

7.2 Greatest common divisor and the algorithm

Definition 16 (GCD) For integers a, b not both zero, $\gcd(a, b)$ is the largest integer dividing both. Two integers are **coprime** when $\gcd(a, b) = 1$.

The Euclidean algorithm computes the GCD by repeatedly replacing the larger number by a remainder — a shrinking, partition-like reduction that must terminate because remainders strictly decrease.

Theorem 17 (Euclidean algorithm) If $a = bq + r$ then $\gcd(a, b) = \gcd(b, r)$. Iterating, the last non-zero remainder equals $\gcd(a, b)$.

Worked example — $\gcd(252, 105)$.

$$252 = 2 \cdot 105 + 42,$$

$$105 = 2 \cdot 42 + 21,$$

$$42 = 2 \cdot 21 + 0.$$

The last non-zero remainder is 21, so $\gcd(252, 105) = 21$.

7.3 Bézout's identity

Running the algorithm **backwards** expresses the GCD as an integer combination of the originals — a fact that decides invertibility and solvability later.

Theorem 18 (Bézout's identity) For integers a, b not both zero there exist integers u, v with

$$au + bv = \gcd(a, b).$$

In particular a and b are coprime iff $au + bv = 1$ for some integers u, v .

Worked example — Bézout for 252 and 105 . Reverse the chain above:

$$21 = 105 - 2 \cdot 42 = 105 - 2(252 - 2 \cdot 105) = 5 \cdot 105 - 2 \cdot 252.$$

So $u = -2$, $v = 5$ give $252u + 105v = 21 = \gcd(252, 105)$.

7.4 Linear Diophantine equations

Theorem 19 (Linear Diophantine equations) The equation $ax + by = c$ has integer solutions if and only if $\gcd(a, b)$ divides c . If so, and (x_0, y_0) is one solution, then with $d = \gcd(a, b)$ **all** solutions are

$$x = x_0 + (b/d)t, \quad y = y_0 - (a/d)t, \quad t \in \mathbb{Z}.$$

Common pitfall Always test the divisibility condition $d \mid c$ **first**. $6x + 4y = 5$ has no integer solution at all, since $\gcd(6, 4) = 2$ does not divide 5 — the left side is always even. Looking for solutions before checking solvability wastes effort on the impossible.

Exercises

20. Compute $\gcd(1071, 462)$ by the Euclidean algorithm and express it as a Bézout combination.
21. Solve $15x + 6y = 9$ over the integers, or prove it has no solution; if solvable, give the full family of solutions.
22. Prove that consecutive integers n and $n + 1$ are always coprime by exhibiting a Bézout combination.

8 Modular arithmetic

The residue classes of Chapter 4 become an **arithmetic** — a finite number system $\mathbb{Z}/n\mathbb{Z}$ where you can add, multiply, and sometimes divide. This is the algebraic home of cyclic phenomena (clocks, days of the week) and the setting in which the previous chapter's GCD decides what is invertible.

8.1 Congruence and its arithmetic

Definition 20 (Congruence) For $n \in \mathbb{N}^*$, write $a \equiv b \pmod{n}$ when $n \mid (a - b)$, i.e. a and b leave the same remainder on division by n . Congruence **respects** addition and multiplication: if $a \equiv a'$ and $b \equiv b'$ then $a + b \equiv a' + b'$ and $ab \equiv a'b'$. Hence arithmetic is well defined on the classes $\mathbb{Z}/n\mathbb{Z} = \{[0], [1], \dots, [n - 1]\}$.

Worked example — Clock arithmetic. Modulo 12, $9 + 5 \equiv 14 \equiv 2$: five hours after 9 o'clock is 2 o'clock. The class $[14]$ and the class $[2]$ are the same block of the partition.

8.2 Invertibility

In $\mathbb{Z}/n\mathbb{Z}$ you cannot always divide. The exact condition for an element to have a multiplicative inverse is, pleasingly, the coprimality of Chapter 7.

Theorem 21 (Invertibility in $\mathbb{Z}/n\mathbb{Z}$) A class $[a]$ is **invertible** — there is $[x]$ with $ax \equiv 1 \pmod{n}$ — if and only if $\gcd(a, n) = 1$. When it exists, the inverse comes straight from Bézout: writing $au + nv = 1$, the relation $au \equiv 1 \pmod{n}$ gives $[a]^{-1} = [u]$.

Worked example — Inverting 5 modulo 12. $\gcd(5, 12) = 1$, so 5 is invertible. Bézout: $5 \cdot 5 - 2 \cdot 12 = 1$, so $5 \cdot 5 \equiv 1 \pmod{12}$ and $[5]^{-1} = [5]$. Indeed $25 \equiv 1$.

Common pitfall Non-coprime classes have **no** inverse and behave strangely: modulo 12, $[3] \cdot [4] = [0]$ even though neither factor is $[0]$. Such **zero divisors** mean you cannot cancel freely — $ax \equiv ay$ does **not** imply $x \equiv y$ unless a is invertible.

8.3 Linear congruences

Proposition 22 (Solving $ax \equiv c \pmod{n}$) The congruence $ax \equiv c \pmod{n}$ is solvable iff $\gcd(a, n)$ divides c . When $\gcd(a, n) = 1$ the solution is unique modulo n , namely $x \equiv a^{-1}c$; in general there are exactly $\gcd(a, n)$ solutions modulo n .

Worked example — A unique solution. Solve $5x \equiv 3 \pmod{12}$. Since $[5]^{-1} = [5]$, multiply both sides: $x \equiv 5 \cdot 3 = 15 \equiv 3 \pmod{12}$. Check: $5 \cdot 3 = 15 \equiv 3$. ✓

Exercises

- 23. Build the multiplication table of $\mathbb{Z}/5\mathbb{Z}$ and identify the inverse of each non-zero class.
- 24. Determine whether $[8]$ is invertible modulo 15; if so find its inverse via Bézout.
- 25. Solve $6x \equiv 9 \pmod{15}$, listing all solutions modulo 15, and explain using $\gcd(6, 15)$ why there is more than one.

9 Pre-stochastic bridge I: partition-and-condition

Here the threads converge. The partition of Chapter 4 and the weighted counting of Chapter 6 combine into the single most useful move in probability — **split into cases, weight each case, recombine** — met now without a single probability axiom, purely as arithmetic over fractions of a finite population.

9.1 Weighted averages over a partition

Definition 23 (Weighted average over a partition) Partition a finite population into classes $A_1, \dots, A_k$ holding fractions $p_1, \dots, p_k$ of the whole (so $p_i \geq 0$ and $\sum_i p_i = 1$). If a quantity takes the value v_i on A_i , its **overall average** is the weighted average

$$\bar{v} = \sum_{i=1}^k p_i v_i.$$

Worked example — Average grade across two sections. Section A (60% of students) averages 14; section B (40%) averages 11. The cohort average is not $(14 + 11)/2 = 12.5$ but the **weighted** $0.6 \cdot 14 + 0.4 \cdot 11 = 8.4 + 4.4 = 12.8$. The larger section pulls the average toward its own.

Remark This is the **law of total expectation** in disguise: the overall average equals the average of the within-class averages, weighted by class size. Next semester the “fractions” become probabilities and the “within-class averages” become conditional expectations, but the arithmetic is identical to what you just did.

9.2 Sequential conditioning on a tree

When conditions are applied in **stages**, a tree records them. Each branch carries the fraction of its parent group having the next property; the fraction of the whole population following a full path is the **product** of the branch fractions along it, and the fraction with a final property is the **sum** over all paths reaching it.

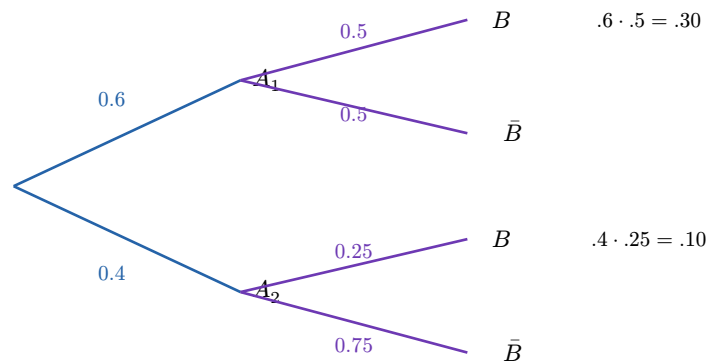


Figure 3: Sequential conditioning. The population first splits by A_1 vs A_2 , then each part splits by B vs $\bar{B}$. The fraction in both A_1 and B is the **product** along the top path, $0.6 \times 0.5 = 0.30$. Summing the two paths that reach B gives the total fraction with property B : $0.30 + 0.10 = 0.40$.

Worked example — Reading the tree. With the branch fractions shown: the fraction of the population in class A_1 and property B is $0.6 \times 0.5 = 0.30$; in A_2 and B it is $0.4 \times 0.25 = 0.10$. The total fraction having property B is $0.30 + 0.10 = 0.40$. Going the **other** way — “of those with B , what fraction came from A_1 ?” — gives $0.30/0.40 = 0.75$, the seed of what next semester is called **Bayes’ rule**.

Common pitfall Branch fractions are **conditional**: the 0.5 on the top branch is the fraction of A_1 that has B , not the fraction of the whole population. Multiply along a path before you compare across paths; adding conditional fractions directly is the most common error here.

Exercises

26. A factory draws 70% of parts from supplier X (2% defective) and 30% from supplier Y (5% defective). Draw the tree and compute the overall defective fraction.
27. For the previous problem, of the defective parts, what fraction came from supplier Y? Identify which path-products you divided.
28. Three urns hold different fractions of red balls; an urn is chosen with given weights and a ball drawn. Set up the weighted average for “fraction red” and evaluate it.

10 Pre-stochastic bridge II: finite dynamics

The final bridge connects this course to both the matrices of MAT11-1 and the Markov chains of S2. A system that hops between finitely many states, one step at a time, is described by a matrix; iterating the system is **raising that matrix to a power**.

10.1 States and the transition matrix

Definition 24 (Transition matrix) A finite-state system with states $1, \dots, m$ is described by a **transition matrix** T whose entry T_{ij} is the fraction of state j that moves to state i in one step. Each column sums to 1 (everything must go somewhere), so T is **column-stochastic**.

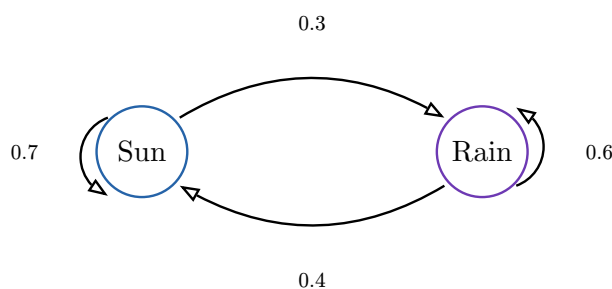


Figure 4: A two-state weather system. A sunny day stays sunny with fraction 0.7 and turns rainy with 0.3; a rainy day stays rainy with 0.6 and clears with 0.4. The arrows out of each state sum to 1 — the columns of its transition matrix.

Worked example — The weather matrix and one step. Ordering states (Sun, Rain), the diagram gives

$$T = \begin{pmatrix} 0.7 & 0.4 \\ 0.3 & 0.6 \end{pmatrix},$$

whose columns sum to 1. If today is certainly sunny, $x_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, then tomorrow's distribution is

$$Tx_0 = \begin{pmatrix} 0.7 & 0.4 \\ 0.3 & 0.6 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.7 \\ 0.3 \end{pmatrix}.$$

10.2 Iterating: matrix powers

Proposition 25 (\$n\$-step transitions) If x is the column vector of current state fractions, then Tx gives the fractions after one step and

$$T^n x$$

the fractions after n steps. The matrix powers are exactly those of MAT11-1, now read as “evolve the system n times.”

Worked example — Two days out. From $x_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, two steps give

$$T^2 x_0 = T \begin{pmatrix} 0.7 \\ 0.3 \end{pmatrix} = \begin{pmatrix} 0.7 \cdot 0.7 + 0.4 \cdot 0.3 \\ 0.3 \cdot 0.7 + 0.6 \cdot 0.3 \end{pmatrix} = \begin{pmatrix} 0.61 \\ 0.39 \end{pmatrix}.$$

Iterating further, x_n settles toward a **steady state** x^* with $Tx^* = x^*$ — here $x^* = \begin{pmatrix} 4/7 \\ 3/7 \end{pmatrix} \approx \begin{pmatrix} 0.571 \\ 0.429 \end{pmatrix}$. This fixed vector is the discrete-dynamics cousin of the affine fixed point of Chapter 6, and the **stationary distribution** of a Markov chain in S2.

Remark Everything in this section is literally a Markov chain — a system whose next state depends only on the current one — written in the language of matrices. When MAT21-4 introduces Markov chains with probability axioms, you will already own the computation: build T , apply it, raise it to a power, find its fixed vector.

Exercises

29. A customer is “loyal” or “switched.” Each month 80% of loyal customers stay loyal and 50% of switched customers return. Write T , and starting from all loyal compute the split after two months.
30. Verify that the columns of your T sum to 1 and explain why this must hold for any transition matrix.
31. Find the steady-state vector x^* of the weather matrix by solving $Tx^* = x^*$ together with the constraint that its entries sum to 1.

11 Python micro-labs

Discrete mathematics is unusually friendly to the computer: its objects are finite and can be **enumerated**. These short labs ask you to confirm by machine what you proved by hand, and to compare **exact counting** against **empirical simulation** — the experimental side of the pre-stochastic bridge.

11.1 Enumerate, then simulate

A finite sample space can be listed exactly — for instance with `itertools.product` or `itertools.combinations` — and outcomes satisfying a property **counted**. The same quantity can be **estimated** by drawing many random trials with `random` and recording the frequency. The two should agree, and the empirical frequency should approach the exact count as the number of trials grows — a hands-on preview of the law of large numbers.

The lab pattern.

1. **Enumerate** the finite sample space and count, by exact combinatorics, the outcomes with the property of interest — the **theoretical** fraction.
2. **Simulate** by generating many random trials and recording the **empirical** fraction with the property.
3. **Compare**. Confirm the empirical fraction converges to the exact one as trials increase, and that both match your by-hand count.

Worked example — Two dice summing to 7. Enumerate the 36 equally likely ordered pairs; exactly 6 of them sum to 7, so the exact fraction is $6/36 = 1/6$. Simulating ten thousand rolls and recording how often the sum is 7 yields a frequency close to 0.167, drifting nearer as trials grow. Counting and simulation corroborate each other.

Exercises

32. Enumerate all $\binom{52}{5}$ poker hands (or a tractable analogue) and count those that are “all one suit”; compare the exact fraction to a simulation.
33. By enumeration, count the permutations of $\{1, \dots, 5\}$ with no fixed point (derangements); compare to the exact count $D_5 = 44$ and to a simulation.
34. Simulate the two-state weather chain of Chapter 10 for many days from a sunny start, and check the long-run fraction of sunny days against the steady-state $4/7$ you computed.